

- ① 오답.md 어제 오답부터 재풀이
- ② 신규는 공부계획 ID대로
- ③ 채점: solve-exam-prep 로 풀이 대조

떠올릴 개념

- 최대전력 $\Leftrightarrow Z_L = Z_{Th}^*$ (켈레 정합)
- $P_{max} = |V_{Th,rms}|^2 / (4 \cdot R_{Th})$
- rms/peak · ½계수 주의

문항 목차 (총 3문항)

- [1] 10.42 — 최대전력전달 · 모의
- [2] 10.44 — 최대전력전달
- [3] 10.46 — 최대전력전달

— 풀이 —

10.42 The phasor voltage $\mathbf{V}_{ab}$ in the circuit shown in Fig. P10.42 is $300\angle 0^\circ$ V (rms) when no external load is connected to the terminals a, b. When a load having an impedance of $200 - j500 \Omega$ is connected across a, b, the value of $\mathbf{V}_{ab}$ is $156 - j42$ V (rms).

Problems **435**

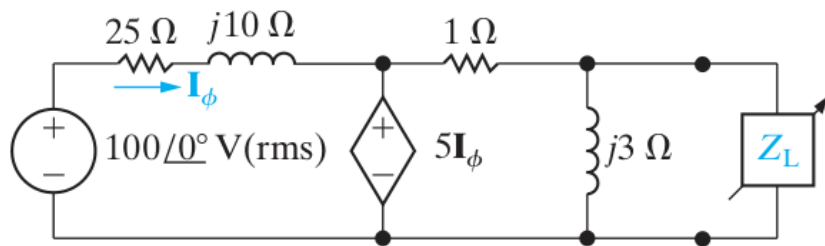
- a) Find the impedance that should be connected across a, b for maximum average power transfer.
- b) Find the maximum average power transferred to the load of (a).
- c) Construct the impedance of part (a) using components from Appendix H if the source frequency is 500 Hz.

Figure P10.42

10.44 The load impedance Z_L for the circuit shown in Fig. P10.44 is adjusted until maximum average power is delivered to Z_L .

- Find the maximum average power delivered to Z_L .
- What percentage of the total power developed in the circuit is delivered to Z_L ?

Figure P10.44



— 풀이 —

10.46 The variable resistor in the circuit shown in Fig. P10.46 is adjusted until the average power it absorbs is maximum.

- Find R .
- Find the maximum average power.
- Find a resistor in Appendix H that would have the most average power delivered to it.

Figure P10.46

